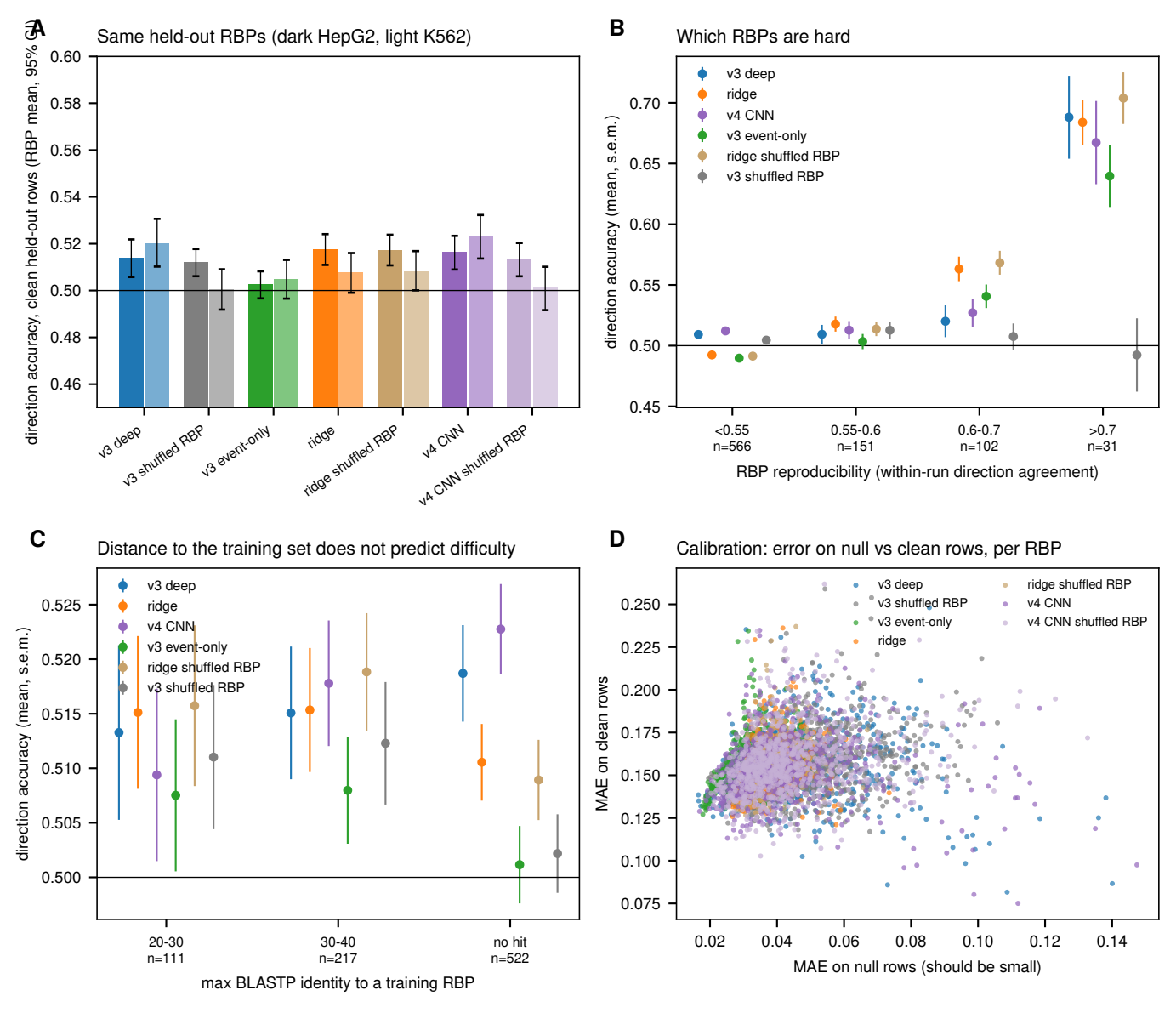

A
Same held-out RBPs (dark HepG2, light K562)
direction accuracy, clean held-out rows (RBP mean, 95% CI)
0.60
0.58
0.56
0.54
0.52
0.50
0.48
0.46
v3 deep
v3 shuffled RBP
v3 event-only
ridge
ridge shuffled RBP
v4 CNN
v4 CNN shuffled RBP
B
Which RBPs are hard
direction accuracy (mean, s.e.m.)
0.70
0.65
0.60
0.55
0.50
0.45
v3 deep
ridge
v4 CNN
v3 event-only
ridge shuffled RBP
v3 shuffled RBP
<0.55
n=566
0.55-0.6
n=151
0.6-0.7
n=102
>0.7
n=31
RBP reproducibility (within-run direction agreement)
C
Distance to the training set does not predict difficulty
direction accuracy (mean, s.e.m.)
0.525
0.520
0.515
0.510
0.505
0.500
v3 deep
ridge
v4 CNN
v3 event-only
ridge shuffled RBP
v3 shuffled RBP
20-30
n=111
30-40
n=217
no hit
n=522
max BLASTP identity to a training RBP
D
Calibration: error on null vs clean rows, per RBP
MAE on clean rows
0.250
0.225
0.200
0.175
0.150
0.125
0.100
0.075
v3 deep
v3 shuffled RBP
v3 event-only
ridge
ridge shuffled RBP
v4 CNN
v4 CNN shuffled RBP
0.02
0.04
0.06
0.08
0.10
0.12
0.14
MAE on null rows (should be small)